

Problem 1. Let $ABCD$ be a cyclic quadrilateral. Let P, Q, R be the feet of the perpendiculars from D to the lines BC, CA, AB , respectively. Show that $PQ = QR$ if and only if the bisectors of $\angle ABC$ and $\angle ADC$ are concurrent with AC .

Problem 2. Find all surjective functions $f: \mathbb{N} \rightarrow \mathbb{N}$ such that for every $m, n \in \mathbb{N}$ and every prime p , the number $f(m + n)$ is divisible by p if and only if $f(m) + f(n)$ is divisible by p . (Here $\mathbb{N}$ denotes the set of all positive integers.)